

CHEM&121: Introduction to Chemistry

Why This Course, Why You

You're here because chemistry is a required step toward a health career — Nursing, Dental Hygiene, or another health-occupations path. This course exists to build the specific chemistry reasoning health professionals actually use: dosage calculations, molecular behavior, and the math-conversion fluency that shows up constantly in clinical work. This isn't chemistry for its own sake — it's the foundation the rest of your health career gets built on.

Chemistry also puts a spotlight on how comfortable you are with learning itself. This isn't easy for anyone, because it's more than recalling facts and getting things right — it's being comfortable in the space between what you know and what you don't yet. Real understanding comes with time and repetition, and there are no shortcuts. Everything in this syllabus is built around getting you through that process, not around catching you out.

Start Here

If you read one part of this syllabus, read this one.

Item	Details
What it costs	Chemfolio [price TBD] + CHEM&121 Lab Manual [price TBD] (Clark Bookstore) + ALEKS (paid through your course registration).
When things are due	Tuesday through Sunday each carry something — see Weekly Flow below for the exact rhythm. Once you learn it, it repeats every module.
Time to budget	About 10–12 hours a week — most of it on Canvas, plus your 4-hour on-campus lab.
How to reach me	Canvas Inbox only — I don't use email during the term. I reply within 24 hours on weekdays.
The two big ones	Weekly lab attendance (required every week, no exceptions) and the Final TYC (Teach Your Community) Project — 10% of your grade, in place of a final exam.
The AI rule	AI can support your learning — don't let it do the learning for you. See the AI Agreement below for exactly where the lines are.
If you fall behind	No explanation needed. Late work loses 1% per hour after the due date, down to zero after about 5 days — then it can still be submitted for zero credit. See Late Work below for the full policy.

Your grade in one line: Discussions 30% · Labs 30% · Quizzes 20% · ALEKS Homework 10% · Final TYC Project 10%.

(Working draft — the teaching team will confirm final category weights before the term begins.)

Course Terms You'll See

A few abbreviations come up constantly in this course. Here's what they mean, so you're not guessing the first time you see them:

Term	What it means
3-2-1	Your weekly Team Discussion format — named for its posting structure (three problem areas, two areas of interest, one insight from the video key).
OOL	Outside Of Lecture — a separate weekly class discussion, not the 3-2-1.
TYC	Teach Your Community — your Final Project, replacing a final exam.
ALEKS	Your online adaptive homework and textbook-access platform.
DAC	Disability Access Center — Clark's office for accommodations.
Chemfolio	Your required lecture/discussion workbook — you buy this!
TL;DR	"Too long; didn't read" — a quick plain-language summary. Wherever you see TL;DR, that line tells you the gist before the official/detailed text below it.

Your First Week

1. Attend your scheduled lab session. Missing it means being dropped from the course.
2. Log into Canvas and read the announcements at the top of the course.
3. Work through Module 0, including the ALEKS/Office/OneDrive setup guide — both need to be done by the end of your first lab, so budget the time.
4. Read this syllabus, especially the Start Here box and the Course Agreements section.
5. Confirm you have both your Chemfolio and lab manual from the Clark Bookstore before class — your first 3-2-1 team post is due that Wednesday.

Course Information

Item	Details
Course Title	CHEM&121: Introduction to Chemistry (Pre-Health)
Course Nbr/Sec	18405/18406, 18407/18408
Quarter/Year	Fall 2026 (September 21 – December 10, 2026)
Location/Time	Lab: STEM Building 222, Mo/We 2:00–3:50 PM (18405) or Mo/We 5:00–6:50 PM (18407). Lecture/discussion: online, asynchronous.
Course Site	Canvas Learning Management System (clarkcollege.instructure.com)
Course Modality	Hybrid — asynchronous online lecture/discussion + biweekly on-campus lab
Credit Hours	5 credits (3.0 hours lecture-equivalent / 4.0 hours lab)
Prerequisites	MATH 96 (grade of "C" or higher), or placement into Math level 45.
Course Description	What the catalog says: Topics in general chemistry applicable to students seeking a 2-year degree in the health-occupations fields. Unit-factor method is applied to problem solving. Topics covered include units of measurement, atomic structure, chemical bonding, energy, the mole concept, nomenclature of inorganic compounds, writing and balancing equations, properties of gases, solutions and colloids, reaction rates and equilibrium, acids, bases and salts, radiation

Item	Details
	and health. What that means in practice: You'll learn to think like a health professional needs to think about chemistry — using math and unit conversions to solve real clinical-style problems, not just memorizing facts. Expect this course to feel like a math class and a foreign-language class rolled together: we use conversions from day one, every week, in both discussion and lab.

Welcome to Class!

TL;DR: No lectures — everything happens through weekly team discussions and hands-on labs. Show up, stay engaged, and ask for help before you're stuck.

Welcome, welcome, welcome! I'm Dr. B (Dr. Karl Bailey, he/him), and I'm so glad to be your Chemistry instructor this term. However you landed in this course, I'm glad you're here — let's get into how it actually runs, week to week.

Dr. B's Note: *Most students who stick with the weekly rhythm and stay engaged do well. The difference isn't talent, it's showing up consistently. Give yourself a fighting chance.*

What preparation/skills do I need for this class?

CHEM is a math class combined with a foreign language course, and we move fast. We use math conversions from day one and keep using them every week, in both discussion and lab. If your math skills need more than a light dusting-off, consider brushing up before or alongside this course — most students taking CHEM&121 find the math the hardest part, not the chemistry itself.

What can I expect in class?

This course is entirely discussion-based online — there are no traditional lectures. Instead, you'll work through weekly Activities from your Chemfolio in a 3-2-1 Team Discussion format and apply what you're learning hands-on in your required weekly on-campus labs. Labs involve real data collection and analysis, not cookbook steps — more on exactly how in Weekly Flow and the Grading section below.

Dr. B's Note: *If you've taken a science course before, don't assume this one works the same way: there are no traditional lectures, quizzes happen in person during lab, and labs are real science, not a recipe to follow. More on exactly how below.*

What if I need an accommodation?

Clark College and the Disability Access Center (DAC) office assist students with disabilities in pursuing their educational goals. DAC can provide classroom accommodations, as well as serve as a resource to the campus community in striving to make Clark College both an accessible and hospitable place for students with disabilities. Interested in getting accommodations? Begin with viewing the steps for requesting accommodations at clark.edu/dac.

Weekly Flow

TL;DR: Something is due nearly every day, Tuesday through Sunday — same pattern every week once you learn it.

Canvas Modules are your home base. This rhythm repeats every module — once you know this table, you know the whole term.

Day	What's due
Monday	New module opens
Tuesday	OOL (Outside Of Lecture) discussion post due
Wednesday	First 3-2-1 Team discussion post due
Thursday	OOL replies due
Friday	Lab Intro upload due
Sunday	Second 3-2-1 Team post + replies to all 3 teammates, ALEKS homework, OOL template upload, Lab Results & Analysis, and Lab Discussion & Conclusions template upload — all due

Attend lab both days assigned in your schedule.

Quizzes are the one exception to this weekly pattern — they're in-person during lab, tentatively Weeks 3, 5, 6, 9, and 11 only.

Dr. B's Note: *Prioritize content learning in your Chemfolio with your team first. Labs are based on the assigned content from the module.*

About Your Instructor

TL;DR: Canvas Inbox only, no email — I reply within 24 hours on weekdays.

Item	Details
Instructor Name and pronouns	Dr. Karl Bailey, Ph.D. (he/him)
How I like to be addressed	Dr. B
Communication	Canvas Inbox only — I do not use email during the term. Not a current student? Reach me at kbailey@clark.edu or 360-992-2313.
Student Hours	Monday & Wednesday, 1:00–2:00 PM and 4:00–5:00 PM (1 hour before each lab section)
Preferred Mode of Communication	Canvas Inbox — I reply within 24 hours on weekdays.

What can you expect from me?

I've spent more than 25 years building and refining active-learning chemistry courses putting students and their learning at the center of my teaching whether in person or online. I've designed everything in this course with you in mind based on what my students tell me works. The first big part of that is learning to struggle when learning anything new. It's normal. So

stick with it and you'll find your footing and confidence with it. Reach out for help directly before things get away from you. I respond to Canvas messages within 24 hours on weekdays.

"Our job as educators is not to prepare you for the real world. You already live in the real world. Our job is to empower you to change it." — Dr. B

What Course Materials Will I Need?

TL;DR: Chemfolio + your lab manual (Clark Bookstore) + ALEKS (set up automatically in Canvas).

Item	Details
Texts/Other learning materials	Chemfolio (lecture workbook) — required, drives our 3-2-1 discussion format. <i>[edition/ISBN/price TBD]</i> CHEM&121 Lab Manual (lab manual) — required. <i>[edition/ISBN/price TBD]</i> ALEKS — required online homework platform, includes access to the Smith textbook. Paid through your course registration.
Companion Web Site	ALEKS (aleks.com), accessed through Canvas
Software	Microsoft Office (primarily Word and Excel) — accessible right inside Canvas through the Microsoft Education integration, using your free Clark student Office 365 account.
Electronic Resources	Canvas Learning Management System, with Microsoft OneDrive linked directly through Canvas's built-in Microsoft Education integration (no separate office.com setup needed). A heads up: this embedded Office/OneDrive connection tripped up a number of students last term. Module 0 has a step-by-step setup guide — work through it in week one, and if you hit an error, contact Tech Hub (techhub@clark.edu · 360-992-2010) rather than troubleshooting alone.
Privacy related to course tools and materials	See Clark's eLearning accessibility and privacy policies page (clark.edu/academics/eLearning) for Canvas, ALEKS, and other tools used in this course.
Accessibility related to course tools and materials	See Clark's eLearning accessibility and privacy policies page (clark.edu/academics/eLearning) for accessibility statements on Canvas, ALEKS, and other course tools.
Where to find necessary materials	Chemfolio and your lab manual: Clark College Bookstore, <i>[bookstore link pending]</i>. ALEKS: access is set up automatically in Canvas once your registration is processed.

What Are the Course Agreements?

TL;DR: Message me in Canvas Inbox, cite anything that isn't yours, never let AI write your discussion posts, and show up ready to work.

Communication

Canvas Inbox is our channel — for everything. I don't use email during the term, so if you send a Clark email, expect it to go unanswered; message me through Canvas Inbox instead. I respond within 24 hours on weekdays.

Citing Your Sources

Use a citation to credit ANY source of information used in your work, be it a website, image, or content that YOU did not come up with. Yes, it's a pain, but it helps when posting content for others who need to trust and rely on it. To avoid plagiarism (when you intentionally or unintentionally present someone else's work as your own), you will need to signal that you are referencing outside material and give credit. Clark College Libraries' plagiarism guide is a handy resource for learning about what plagiarism is and how to avoid it. I am here if you have questions about how to correctly cite outside information in your work.

AI (Artificial Intelligence) Guidelines — Community AI Agreement

TL;DR: Never use AI to write anything you submit in discussions — no exceptions. AI can help you double-check your own lab/ALEKS work if you declare it. Quizzes and the Final Project: no AI, period.

This agreement is a community commitment built on honesty, growth, and professionalism. AI is a more powerful tool than simply an advanced Google search. Learn to leverage it wisely, use it responsibly, and declare it when submitting your work. By enrolling in this course you are agreeing to abide by this agreement.

For discussions specifically: no AI-generated content is ever allowed in anything you submit — no exceptions, and declaring it doesn't change that. The 3-2-1 format, OOL posts, and Team Lead summaries all require your own reasoning, in your own words. AI-generated discussion content will not receive credit, whether declared or not.

Four Principles

- **Caution** — AI can be wrong. Double-check with class notes, labs, and trusted sources.
- **Trust** — Honesty builds our learning community. Declaring AI use has no penalty. Hiding it does.
- **Relevance** — Use AI to support learning, not to skip actually doing the work.
- **Transparency** — Just declare it. Add "This submission used AI." at the end of your lab answer or ALEKS work.

By Assignment

- **Discussions** — No AI-generated content anywhere in your posts, replies, and includes the LEADER CHECK IN reply. All posts and replies must be written by the student, with no exception. AI cannot watch a video for you, work through an activity's objectives, or learn for you. AI cannot produce reflective thoughts for a human being.
- **Labs** — OK to help interpret or reflect on your own data and results. Never make up, falsify, or use data that is not your own. Attribute AI use in all lab submissions.
- **ALEKS Homework** — OK to support learning. Don't just grab answers — this is exactly the category where leaning on AI defeats the purpose, which is why it's weighted lowest in your grade.
- **Quizzes & Final TYC Project** — No AI allowed, at any level.

Follow-Up

- **Discussions** — AI-generated content in any post, reply, or Team Lead summary = zero credit for that submission, regardless of declaration. Repeated instances = academic integrity referral.
- **Labs** — fake data = zero for the lab. AI answer without declaration = zero for the question. No regrades.
- **Quizzes/Final Project** — suspected use = zero points and academic integrity referral.

REMINDER: Posting course materials online anywhere — including uploading Canvas pages, Activity content, or video keys to AI tools — is academic dishonesty under the Clark College student code of conduct. Course materials are provided for your learning, not to train or prompt AI on your behalf.

Dr. B's Note: *You're entering health careers where accuracy and integrity matter. AI is here to stay — but how you use it says everything about your professionalism. Own it, declare it, and use it to grow.*

Classroom Norms

- Complete each week's assignments — discussions, labs, ALEKS — by the posted due date.
- Attend your scheduled labs (both) every week; all lab hours are required to pass the course (all labs with combined score >60%).
- Be a courteous member of the team and don't get in the way of anyone else's learning: post and reply before the due dates, not hours or minutes before.
- Academic and intellectual honesty are expected at all times (act professionally now, not when you get the job).
- Tell me as early as you can about anything affecting your ability to take part fully.

"How you do anything is how you do everything."

What Are the Learning Goals in This Course?

TL;DR: By the end, you can apply chemistry reasoning to real health-career problems — not just recall facts.

Student learning outcomes are the skills and abilities that you are working to acquire, practice, and apply in class. They represent the learning goals for the term.

Upon successful completion of this course, students will be able to:

- **CLO1:** Applies the fundamental rules behind element reactivity as it applies to how bonds are formed and broken, with context to physiology, pharmacology, and metabolism.
- **CLO2:** Demonstrates a clinical-based level of understanding of the forces at play within and between molecules and ions, and their impact on whether substances are water (hydrophilic) or fat (hydrophobic) soluble.
- **CLO3:** Displays critical thinking and mathematical skills to solve contextual problems necessary in the treatment of patients as a health care professional.

These outcomes are measured through your weekly 3-2-1 Team Discussions, lab reports (Results & Analysis and Discussion & Conclusions), in-lab quizzes, and the Final TYC Project. Rubrics for each are posted in the relevant Canvas module.

How Will I Be Graded, and Why?

TL;DR: Discussions 30% · Labs 30% · Quizzes 20% · ALEKS 10% · Final TYC Project 10%. Here's why those weights, and the full breakdown.

These weights aren't arbitrary. Labs and Discussions carry the most weight because that's where the real demonstration of your learning happens — applying what you know, not just recognizing it. ALEKS is weighted lightly because it's practice, not proof of understanding. The Final TYC Project replaces a comprehensive exam because teaching a concept to someone else is a better test of whether you actually understand it than a timed exam is (e.g. "Learn on, do one, teach on.")

Assignment Categories and Their Percentage of the Final Grade

This is a working draft for Fall 2026 — the teaching team will confirm final weights before the term begins.

Category	Weight	What it is
Discussions	30%	Weekly 3-2-1 Team Discussions + OOL post and replies
Labs	30%	Weekly on-campus labs, including INTRO data collection, the RESULTS+ANALYSIS and DISCUSSION+CONCLUSION Canvas assignments
Quizzes	20%	In-person, given during lab (tentatively Weeks 3, 5, 6, 9, and 11)
ALEKS Homework	10%	Online adaptive homework for content practice
Final TYC Project	10%	Teach Your Community project — no final exam – details in the TYC Canvas module

About the Lab Requirement

Lab hours are part of this course's official credit hours — 1 of your 5 credits is earned specifically through lab. Because of that, each lab must genuinely demonstrate your work, not just check a box.

- Each lab has three parts: INTRO (your uploaded lab notebook pages), Results & Analysis (a Canvas quiz-type assignment, including Excel graphing), and Discussion & Conclusions (a Word template where you replace the provided language with your own analysis, submitted to Canvas and TurnItIn).
- Each individual lab must score above 60% combined across its three parts — this is not an average across all labs, it's per lab.
- A lab counts as "missed" toward your credit-hour requirement if you don't attend/submit it, or if it scores below 60%. You cannot miss more than about two labs and still earn the lab credit for this course.

Grading Scale

Note: Clark does not award C-, D+, or D- in this course — a C- does not transfer, so grading stops at a clean C, then D, then F.

Percent	Grade	Percent	Grade	Percent	Grade
93 up to 100	A	83 up to 86	B	72 up to 76	C
90 up to 92	A-	80 up to 82	B-	63 up to 71	D
87 up to 89	B+	77 up to 79	C+	0 up to 62	F

Late Work

Life happens during an academic term. I trust you know your situation and I don't need an explanation. My position is to support your learning and your success.

- All assignments remain open after the due date, losing 1% per hour until worth zero points (approximately 5 days). Assignments can still be submitted after this, but the score will be zero.
- In-person lab attendance and the Final TYC Project have no grace period.
- No due dates extend beyond the course end date.
- If something requiring a prolonged absence occurs, contact me via Canvas Inbox and let's see what solutions are possible.

Feedback

Discussions and other work are graded within 48 hours of each due date.

Revision

There is no separate revision or regrade policy at this time — the late-work window above functions as the built-in opportunity to catch up before work is scored.

What Resources and Support Services Are Available to Me?

TL;DR: Tech Hub for Canvas/software problems, Tutoring or Librarians for coursework help, Basic Needs Hub if life is making this term hard.

Technical Support

Item	Details
Tech Hub	Canvas, software, or account trouble — Scarpelli Hall 135 · techhub@clark.edu · 360-992-2010
To report a Canvas problem	The Help button in Canvas → "Report a Problem" (replies come to your Clark Gmail)
Setup help	Module 0 has the Office/OneDrive setup guide — this step trips people up, so budget real time and reach out to Tech Hub if you hit an error

Academic Support

Item	Details
Student Success Coaches	One-on-one coaching for study/organization skills, emotional and personal support, and connections to college resources. PUB 002 or Zoom.
Clark College Tutoring Services	In-person and virtual tutoring via Zoom; eTutoringOnline.org for additional online writing lab support.
Clark College Librarians	One-on-one help with research projects and source citations, in person or by chat.

Student Support

Item	Details
Counseling and Health Center	Physical and mental health services to support student wellness and academic success.
Office of People and Culture (OPC)	Supports systemically non-dominant groups navigating Clark College; the Diversity Center is a great place to visit between classes.
Penguin Pantry	Monthly pre-packaged food boxes. Contact Cath Busha, Dean of Student Engagement (cbusha@clark.edu).
Basic Needs Hub	Help with food, housing, and financial challenges. basicneeds@clark.edu · 360-992-2766 · Gaiser Hall (GHL) 216.

Item	Details
Advising Services	Co-create academic plans aligned with your career and personal goals.
Career Services	Resources and strategies for career and employment goals. 360-992-2902 · careerservices@clark.edu
Veterans Center of Excellence	Resources, community, and support for U.S. military service members. PUB 015 · 360-992-2073 · Vetresources@clark.edu

You can find all these resources as well as links to the Clark bookstore, C-Tran, campus maps, and campus life and activities at the Clark Students website.

How Should I Get Started in Class?

See Your First Week, near the top of this syllabus, for the exact five things to do before anything else.

Appendix: Official College Policies

Everything below is required Clark College policy language, included in full because it has to be — not because it's what this course is about. A TL;DR sits above each one so you can get the gist and move on, or dig into the full text if it applies to you.

ADA Accommodations

TL;DR: If a disability or temporary health condition affects your coursework, DAC can set up accommodations for this class — contact them as early as you can.

Clark College and Disability Access Center (DAC) assist students with disabilities in pursuing their educational goals. Clark College is committed to assuring that its services are accessible to individuals with disabilities. Disability Access Center (DAC) coordinates accommodations for students with disabilities and/or temporary health conditions (could include a temporary injury or pregnancy). DAC approves and provides student disability accommodations for classes in partnership with faculty. Students with disabilities who believe that they may need accommodations in this class are encouraged to contact DAC as soon as possible to better ensure that accommodations are implemented in a timely manner. All accommodations must first be approved through Disability Access Center. Interested in getting accommodations? You can request accommodations by following steps on the DAC website www.clark.edu/dac. Or contact DAC one of these ways:

- DAC Main Phone: 360-992-2314. Video Phone for ASL: 360-991-0901.
- Email: dac@clark.edu. Fax: (360) 992-2879.
- Physical Location: Penguin Union Building (PUB) 002.

If you have already been approved for accommodations, don't forget to request them every term in myAccess. Students who need to test at DAC also need to schedule their own exams in myAccess or by contacting DAC. Exams should be scheduled with DAC with at least 5 business days' notice. Please reach out to DACtesting@clark.edu if you have any questions about your testing accommodations.

Once you have established accommodations with Disability Access Center, please contact me as your instructor to discuss your needs in this course.

Non-discrimination Policy

TL;DR: Clark prohibits discrimination and harassment, and there's a formal process if you experience it.

The college affirms a commitment to freedom from discrimination for all members of the college community. The responsibility for, and the protection of, this commitment extends to students, faculty, administration, staff, contractors, and those who develop or participate in college programs. It encompasses every aspect of employment and every student and community activity. The college expressly prohibits discrimination on the basis of race, color, national origin, age, perceived or actual physical or mental disability, pregnancy, genetic information, sex, sexual orientation, gender identity, marital status, creed, religion, honorably discharged veteran or military status, citizenship, immigration status, or use of a trained guide dog or service animal. Harassment is a form of discrimination. Any person who believes they have been discriminated against or harassed by Clark College or its employees may file a formal grievance to:

Human Resources Department — 1933 Fort Vancouver Way, Baird Hall 133 — Phone: (360) 992-2105 — Email: hr@clark.edu

Title IX and Sexual Misconduct

TL;DR: If you or someone you know has experienced sexual misconduct, there are people and resources here to help — you're not alone.

Clark College is committed to helping individuals achieve their educational and professional goals and to providing a community that is supportive of students who may have experienced sexual misconduct. The Clark community wants survivors of sexual misconduct, in any form, to know that they are not alone — we are here to help.

If you, or someone you care about, have been impacted by sexual misconduct, the most important thing to know is that you are not alone — there are people who can and want to help. People are available to listen, offer support, provide options, and take actions to help ensure that you and our community are safe. Survivors of sexual misconduct respond to their experiences in different ways. There is no 'right way' to feel or 'right thing' to do. Learn more about campus and community resources, reporting options, Title IX, personal safety, and how to help a friend who may have experienced sexual misconduct at clark.edu/title-IX. You can also contact the Counseling and Health Center if you would like to pursue counseling.

Student Rights and Responsibilities

TL;DR: This covers things like FERPA, absences for reasons of faith, and your general rights as a student — reach out if any of it applies to you.

On Clark's Student Rights and Responsibilities page, you can find information about freedom of access to education, student absence for reasons of faith or conscience, FERPA, emergency and health information, etc. Please get in touch if you would like to discuss absences for reasons of faith. I am here to support you.